Rakt Setu & BloodBank: Project Documentation

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Overview

Two connected websites came out of this work. Rakt Setu is a live, free registry that connects village blood donors with families in an emergency. BloodBank is the 2022 college site, redesigned into an information site that sends visitors to Rakt Setu.

	Rakt Setu (V1)	BloodBank (redesign)
Purpose	Find a nearby willing donor and message them on WhatsApp; donors register and manage themselves	Explain who can donate, the process, blood-group matching and why it matters
Live site	rakt-setugov.netlify.app	tamojeetk.github.io/Blood-Donation-Site (GitHub Pages)
Code	github.com/tamojeetK/rakt-setu	github.com/tamojeetK/Blood-Donation-Site
Built with	Plain HTML/CSS/JS, Netlify Functions, Supabase Postgres	Plain HTML/CSS/JS
Languages	English, Hindi, Bengali	English
Running cost	₹0 (free plans)	₹0

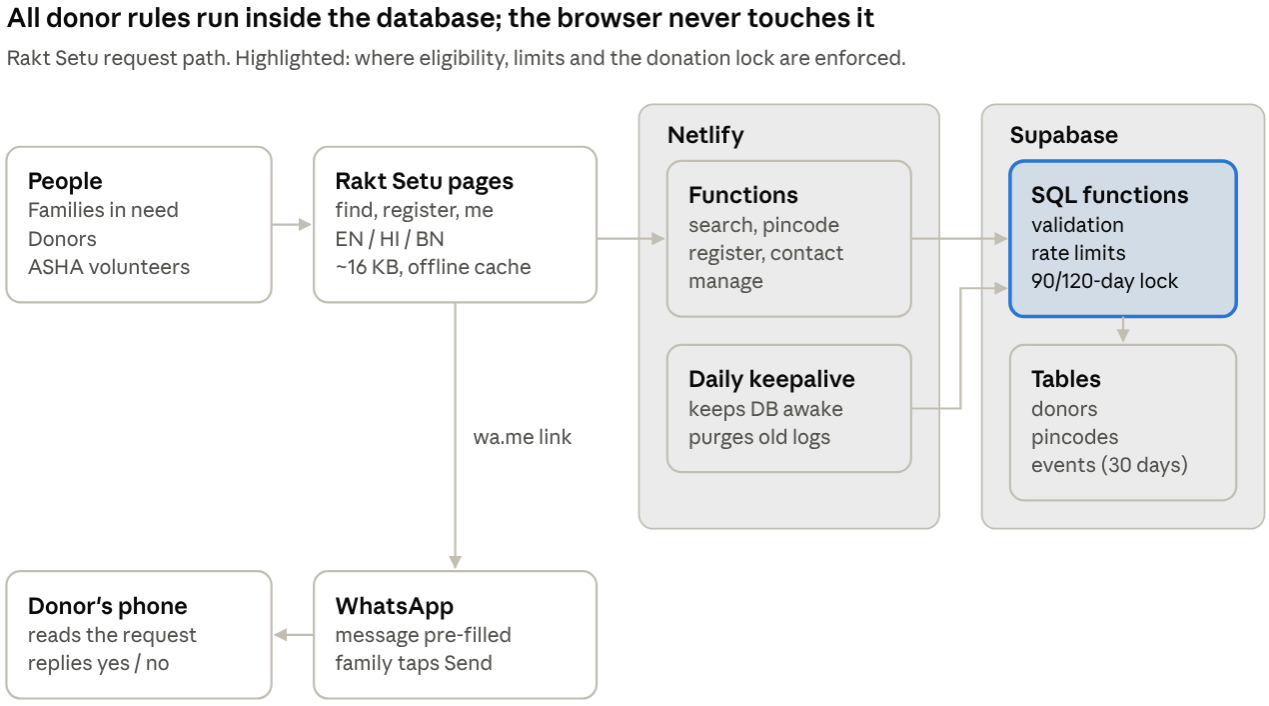
The problem Rakt Setu solves. In rural hospitals, families often have to find "replacement donors" themselves during an emergency. They lose hours phoning around. Rakt Setu gives them a pre-built list of nearby voluntary donors that they can reach in one tap.

Design goals

- Zero running cost, with no paid APIs
- No app download; it works in any phone browser, including on slow 3G
- Donor privacy: phone numbers are never shown publicly
- Follows India's NBTC donor-selection rules, including the 90/120-day gap between donations
- Three languages from day one

Architecture

Rakt Setu has three layers, all on free plans: static pages in the browser, small server functions on Netlify, and a Supabase Postgres database where every rule is enforced. Messages travel over the family's own WhatsApp, so there is no messaging cost.



Rakt Setu architecture · browser, Netlify, Supabase, WhatsApp

The browser only ever talks to the Netlify functions. They hold the secret database key and call one SQL function each. WhatsApp is opened by a normal `wa.me` link, so no WhatsApp API is needed.

Layer	Technology	Free-plan role
Pages	Plain HTML, CSS, JavaScript (`public/`)	Served by Netlify's CDN
Server functions	Node.js (`api/`, wrapped by `netlify/functions/`)	Netlify Functions; daily scheduled function
Database	Supabase Postgres with SQL functions	500 MB database; paused after about a week idle, hence the keepalive
Messaging	WhatsApp deep links (`wa.me`)	Free, uses the family's own WhatsApp
Code	GitHub (`rakt-setu`)	Push to `main` deploys

Layer	Technology	Free-plan role
BloodBank site	Plain HTML/CSS/JS on GitHub Pages	Links into Rakt Setu

How it works

Four kinds of people use Rakt Setu. None of them needs an account or an app.

A family that needs blood

1. Scan the QR code at the hospital, or open the link.
2. Tap **Need blood?**. The page first suggests checking blood-bank stock on e-RaktKosh.
3. Enter a 6-digit pincode (or a district name) and the blood group needed.
4. See up to 30 available donors who can give to that group. Each card shows first name, blood group, area and distance; no phone numbers.
5. Tap **Message on WhatsApp**, enter the hospital name and units needed, and tick "This is a real medical need".
6. WhatsApp opens with a message already written in the donor's chosen language. The family presses Send. **Call instead** opens the phone dialler.

Results are sorted: same pincode, then same district, then within 60 km; donors who confirmed recently first; exact group before compatible groups; then by distance. The order rotates daily so the same donors are not always on top.

A donor

1. Tap **Give blood** and fill in the form: name, WhatsApp number, blood group, gender, year of birth, weight, pincode, village (optional), last donation date and message language.
2. Tick the health self-checks and the consent box.
3. Receive a private link (`/me#<secret>`). **Save link in WhatsApp** sends it to their own chat.
4. Later, open the link to tap **I gave blood today**, pick another date, pause, say "I'm still available", or delete their details.

After a donation the donor is hidden for 90 days (men) or 120 days (women and others), then reappears automatically.

An ASHA worker or youth club volunteer

They register villagers one at a time from their own phone during village meetings, up to 25 sign-ups per phone per day. Each donor's private link should go to that donor's own WhatsApp.

The admin (you)

You see and edit all records in the Supabase dashboard (**Table Editor → donors**). You can correct a donor's details if they phone you, delete a record, and export a CSV backup.

Privacy and safety rules

A donor's phone number leaves the database in exactly one place: the contact request, which is logged and rate-limited. A wa.me link has to contain the number, so the family does see it once they ask. The protection is in who can ask and how often.

Rule	Value	Enforced in
Phone numbers in search results	Never	`search_donors()` does not return them
Needed before a number is released	Hospital name + "real need" tick	`request_contact()` + contact form
Donor chats a family can open	8 per 24 h (re-opening the same donor doesn't count)	`contacts_per_ip_day`
Requests a donor can receive	3 per 24 h, then hidden for the rest of that day	`contacts_per_donor_day`
Gap after donating	90 days men, 120 days women and others	`gap_days()` / `is_available()`
Age / weight to register	18–65 years / at least 45 kg	database + form
Donor not confirmed for 6 months	Shown lower, labelled "Not confirmed recently"	`stale_days` = 180
Donor not confirmed for 12 months	Hidden from search	`hide_days` = 365
Sign-ups from one phone	25 per 24 h	`registers_per_ip_day`

Rule	Value	Enforced in
Visitor IP addresses	Never stored; only a salted hash	`api/_lib.js`
Request and registration logs	Deleted after 30 days	`daily_maintenance()`
Donor's private link	Stored only as a SHA-256 hash	`manage_token_hash` column
Browser access to the database	None; tables locked with row-level security	end of `db/schema.sql`
Bot sign-ups	Hidden "website" field (honeypot) rejects bots that fill it	`register.js`, `contact.js`

All numeric limits live in one function, `cfg()`, at the top of `db/schema.sql`. Change a number there and re-run the file to apply it.

Consent. Registration requires ticking: "I agree that my first name, area and blood group can be shown to families looking for blood, and my number shared with a family when they ask to contact me. I can pause or delete my details any time." The time of consent is stored in `consent_at`. A plain-language privacy page is at `/privacy` in all three languages.

Health self-check. Donors confirm they are well today; have had no tattoo, piercing or major surgery in 12 months; have no heart disease, cancer, hepatitis, HIV or bleeding disorder; and (women) are not pregnant, breastfeeding or within 12 months of delivery. The blood bank still does the full medical screening.

Database

All data and business rules live in one Supabase Postgres database, created by `db/schema.sql`. The file is safe to re-run: tables are created only if missing, and functions are replaced.

Tables

Table	One row per	Key columns
`donors`	Registered donor	`id`, `name`, `phone` (unique, 10 digits), `blood_group`, `sex` (M/F/O), `birth_year`, `weight_kg`, `pincode`, `district`, `state`, `locality`, `lang` (en/hi/bn), `last_donated_on`, `paused`, `confirmed_at`, `consent_at`, `manage_token_hash`, `created_at`
`pincodes`	Indian pincode (19,300 rows)	`pincode`, `district`, `state`, `lat`, `lon`
`events`	Registration or contact request (kept 30 days)	`kind` (register/contact), `ip_hash`, `donor_id`, `at`, `detail` (hospital, units, group)

The pincode data is India Post's All India Pincode Directory from data.gov.in (Government Open Data License – India), via the IndiaPIN package's December 2021 snapshot. It is bundled as `db/pincodes-india.csv`. Pincodes without coordinates borrow their district's centre. Some coordinates are approximate, so distances are a guide only.

Functions

Function	What it does
`register_donor(p, ip_hash)`	Validates every field, checks rate limit and duplicates, fills district/state from the pincode table, saves the donor, returns the one-time private link token
`search_donors(place, group, limit)`	Returns available compatible donors near a pincode or district, sorted as described above; no phone numbers
`request_contact(donor, ip_hash, p)`	Checks hospital name, availability and both rate limits, logs the request, returns phone + first name + language
`manage_donor(token, action, date)`	Private-link actions: get, pause, resume, donated, confirm, delete
`lookup_pincode(pincode)`	Returns district and state for the registration form

Function	What it does
`daily_maintenance()`	Deletes logs older than 30 days; also keeps the free project awake
`is_available(donor)`	Not paused, past the donation gap, aged 18–65, confirmed within 12 months
`compatible_donors(group)`	Red-cell compatibility: e.g. A+ patient → A+, A−, O+, O− donors
`cfg(key)`	All tunable limits in one place

Only the server key (`service_role`) can call these functions. The `anon` and `authenticated` roles that a browser would use are refused for both the tables and the functions.

API endpoints

Six small server functions sit between the browser and the database. They live in `api/` and run on Netlify through thin wrappers in `netlify/functions/` ; the same files also run on Vercel unchanged. They do light input cleaning, hash the visitor's IP, and call one database function each.

Endpoint	Method	Input	Returns
`/api/search`	GET	`place` (pincode or district), `group`	`{donors:[{id, first_name, blood_group, exact, area, distance_km, tier, stale}]}`
`/api/pincode`	GET	`p` (6 digits)	`{found, pincode, district, state}` (cached 24 h)
`/api/register`	POST	donor form fields + `consent` , `healthy`	`{token, next_eligible}`
`/api/contact`	POST	`donor_id` , `hospital` , `units` , `from` , `group`	`{whatsapp, call}` links
`/api/manage`	POST	`token` , `action` , `date`	donor status, or `{deleted:true}`
`/api/keepalive`	daily schedule	none	runs `daily_maintenance()`

WhatsApp message. `/api/contact` builds the text in the donor's language (`api/_messages.js`). Example in Hindi: "नमस्ते Ram जी, मुझे आपका नंबर Rakt Setu (रक्तदाता रजिस्ट्री) से मिला। Kharagpur SDH में एक मरीज़ को तुरंत A+ खून की ज़रूरत है (2 यूनिट)…" It ends by asking the donor to reply "No" if they can't help, so the family can move on quickly.

Error codes

The database raises a short code; the API maps it to an HTTP status; the page shows a translated message.

Code	HTTP	Shown to the user (English)
`RATE_LIMIT`	429	Too many requests from this phone today
`ALREADY_REGISTERED`	409	This number is already registered
`UNAVAILABLE`	409	This donor is not available right now
`NOT_FOUND`, `BAD_TOKEN`	404	Link not valid, or details deleted
`BAD_PHONE`, `BAD_PINCODE`, `AGE`, `WEIGHT`, `NO_CONSENT`, `NOT_ELIGIBLE`, `NEED_HOSPITAL`, `BAD_DATE`, `BAD_INPUT`	400	A specific fix-the-form message
`SERVER_ERROR`	500	Something went wrong (details only in server logs)

Frontend and languages

The site is plain HTML, CSS and JavaScript in `public/`, with no framework and no build step. A first visit downloads about 16 KB compressed, and the home page loaded in about 1.2 seconds on a simulated slow-3G phone.

Page	Path	Purpose
Home	`/`	"Need blood?" and "Give blood" buttons, e-RaktKosh link, how it works
Find	`/find`	Pincode + blood-group search, donor cards, WhatsApp contact form
Register	`/register`	Donor form with live pincode lookup, health checks, consent, private link

Page	Path	Purpose
My details	`/me#<token>`	Donor's private page: donated, pause/resume, confirm, delete
Privacy	`/privacy`	What is stored, why, who sees what, retention

Languages. Every word on screen is in `public/i18n.js`, in English, Hindi and Bengali. A language switch sits in the header on every page. The first visit picks the phone's language when it is Hindi or Bengali, and the choice is remembered on that device. To fix a translation, edit the string in that file; nothing else changes.

Phones and laptops. Layouts are built for phones first: large tap targets, 16 px side margins, no sideways scrolling at 360 px wide. From 760 px the home cards sit side by side and the contact form opens as a centred dialog. From 960 px the search form stays on the left with results in a grid beside it. On a laptop, WhatsApp opens WhatsApp Web or the desktop app.

Weak networks. A small service worker (`sw.js`) caches the pages, styles and scripts, so they still open when the signal drops. Requests time out after 25 seconds with a clear "No internet" message. If someone types before the script has loaded, the form catches up when it arrives.

Security headers. `netlify.toml` (and `vercel.json`) set a strict Content-Security-Policy, block framing, stop MIME sniffing and send no referrer.

Deployment

Rakt Setu is live on Netlify, connected to the `rakt-setu` GitHub repo, with data in Supabase. Every push to `main` redeploys the site automatically.

Supabase (database), done once

1. Create a free project, region South Asia (Mumbai).
2. **SQL Editor** → paste `db/schema.sql` → **Run**.
3. **Table Editor → pincodes → Insert → Import data from CSV** → `db/pincodes-india.csv` (19,300 rows).
4. Copy the **Project URL** and the **secret key** (`sb_secret_…`, or the legacy `service_role` key).

Netlify (website + functions), current host

1. Sign up with email, then **Add new project → Import an existing project → GitHub → rakt-setu**.
2. Leave build settings empty; `netlify.toml` sets them (publish `public/`, functions in `netlify/functions/`).
3. Add the environment variables below, then deploy.
4. If variables change later: **Deploys → Trigger deploy → Deploy site**.

Environment variables

Name	Required	Value
`SUPABASE_URL`	Yes	Supabase Project URL
`SUPABASE_SERVICE_KEY`	Yes	Supabase secret key. Never put this in website code
`IP_SALT`	Yes	Any long random text, used to hash IP addresses
`APP_NAME`	No	Name used in WhatsApp messages; default "Rakt Setu". (Not `SITE_NAME`, which Netlify reserves.)
`CRON_SECRET`	Vercel only	Any long random text; protects `/api/keepalive`

Vercel (alternative)

The repo also deploys to Vercel unchanged: import the repo, add the same variables plus `CRON_SECRET`, deploy. `vercel.json` sets the output folder, clean URLs, headers and a daily cron. We moved to Netlify because the GitHub account was already linked to an old college Vercel account.

BloodBank site

Served by GitHub Pages from the `main` branch of `Blood-Donation-Site`. Changes go live a minute or two after a push. If it ever stops, check **Settings → Pages → Deploy from a branch → main / (root)**.

Operations

Day to day there is nothing to run. The tasks below come up occasionally.

Task	How	When
See or correct donors	Supabase → Table Editor → `donors`	When a donor asks
Remove a donor who lost their link	Delete their row in `donors` (the link can't be re-sent; it is stored only as a hash)	On request
Back up donor data	`donors` table → ⋯ → Export to CSV; keep it somewhere private	Every few weeks
Keep Supabase awake	Automatic: Netlify runs `keepalive` daily (free projects pause after about a week idle)	Daily, nothing to do
Purge old logs	Automatic: same daily job deletes `events` older than 30 days	Daily, nothing to do
Change a limit (e.g. requests per donor)	Edit `cfg()` in `db/schema.sql`, re-run it in SQL Editor	As needed
Fix a translation	Edit `public/i18n.js` (screens) or `api/_messages.js` (WhatsApp text), push	As needed
Ship any code change	Push to `main`; Netlify redeploys in about a minute	As needed
After changing CSS/JS	Bump `VERSION` in `public/sw.js` so phones fetch the new files	With each change
Check errors	Netlify → Logs → Functions	If users report problems

Pilot launch checklist

- [ ] Rename the Netlify project so its address doesn't contain "gov", then update the links in `Blood-Donation-Site`
- [ ] Have a native speaker review the Hindi and Bengali text
- [ ] Register a test donor, search, open WhatsApp, mark donated, delete
- [ ] Make a QR code for the live link and print it with "Need blood? Scan here / खून चाहिए? स्कैन करें / রক্ত দরকার? স্ক্যান করুন"
- [ ] Put it up at one Community Health Centre or district blood bank
- [ ] Brief ASHA workers and youth clubs on registering villagers
- [ ] Delete the two GitHub access tokens used during setup

Testing

Every flow passed before launch. Tests ran against a local copy of the stack: Postgres 16 with the real schema and pincode data, PostgREST standing in for Supabase's API, and headless Chromium driving the pages. The live Netlify site itself was checked by you after deploying.

Area	What was checked	Result
Database rules	Search ranking, district matching, 3-requests-per-donor hiding, repeat request not counted, pause, donated lock, delete	Pass
Validation	Bad phone, under 18, under 45 kg, duplicate number, no consent	Correct error each time
Lockdown	Browser role reading `donors` or calling functions	Permission denied
API	All endpoints, wrong method (405), bad token (404), keep-alive with and without secret	Pass
Netlify wrappers	Each function called the way Netlify calls it; visitor IP passed through and hashed	Pass
Full journey on a phone	Register in Hindi → search in Bengali → WhatsApp opens with Hindi message → private link → donated → delete	Pass, no script errors
Donation lock	Woman donating 7 Nov 2026 hidden until 7 Mar 2027 (120 days)	Pass
Slow network	Simulated slow 3G on a Pixel 5	Home page about 1.2 s, about 20 KB
Layout	No sideways scroll at 360, 800, 1024 and 1440 px	Pass
Pincode import	19,300 rows, all with coordinates	Pass

Three bugs were found and fixed during testing:

- typing before the script loaded on slow networks skipped the pincode lookup
- a 14-character limit cut off "+91 99887 76655"

- district names differed between the donor form and the postal data ("Paschim Medinipur" vs "Medinipur West")

The tests are kept in `test/` (`sql_test.sql`, `e2e.cjs`, `dev-server.mjs`).

BloodBank site redesign

The 2022 college site (Group 1, section K20KP, Lovely Professional University: Dev Sharma, Mridul, Abhishek Kumar, Tamojeet Kuila) was rebuilt from scratch in November 2026. The original files remain in the repo's git history.

Before and after

	2022 version	2026 redesign
Structure	14 HTML pages (several unused drafts) and 13 stylesheets	One page with sections and a sticky menu (5 files)
Phones	Only 2 of 13 stylesheets adapted; menu overlapped the text	Built for 360 px phones up to wide desktops
Forms	Login, register, find, feedback with no backend	Removed; finding and registering go to Rakt Setu
Size	15 MB (one 8.2 MB photo)	About 31 KB plus web fonts
Broken parts	Missing images and CSS, case-sensitive link, stacked login toggle	None found in testing
Contact details	Placeholder addresses and a real personal phone number	Removed; emergency 108 and e-RaktKosh instead
Look	Light, single-column	Light and dark mode, keyboard accessible

What the new page contains

1. **Hero** with "Find a donor near you" and "Become a donor" buttons (both open Rakt Setu) and an emergency hint.
2. **Key facts**: one donation helps up to 3 patients; 8–10 minutes to donate; 90/120 days between donations; 14 June is World Blood Donor Day.
3. **The donation process** in four steps: register and health history, mini check-up, donation, rest.

4. **Who can give**: six criteria cards, a 6-question "Can I donate today?" checker that runs only in the browser, and expandable waiting periods.
5. **Blood-group matcher**: tap a group to see who it can give to and receive from.
6. **Why donate**, and three myths vs facts.
7. **Share** on WhatsApp or copy the link.
8. **Team and timeline**: 2022 v0 → 2026 redesign → 2026 Rakt Setu.

Health facts corrected

Topic	2022 text	Now (NBTC guidelines)
Minimum weight	50 kg	45 kg
Age	"WHO… 17 to 66"	18–65 years
Blood pressure	Systolic 100–180, diastolic 50–100	100–140 / 60–90 mm Hg
Pulse	50–100	60–100, regular
Gap between donations	"More than 3 months" for everyone	90 days men, 120 days women
Storing your own blood	Suggested for rare groups	Removed (misleading)

The "Find a donor" links point to the Rakt Setu address in two places: `index.html` and the `APP` line at the top of `assets/script.js`.

Known limitations and roadmap

V1 is ready for a one-hospital pilot. These items should be handled before it grows to more districts.

Item	Why it matters	Planned fix
Netlify address contains "gov"	People may think it's a government service	Rename the Netlify project; update links
Phone numbers aren't verified	Someone could register another person's number	V2: donor sends one WhatsApp message to confirm before being listed. For now, ASHA-assisted sign-up

Item	Why it matters	Planned fix
Translations written by AI	Wording may be unnatural or unclear	Review by native Hindi and Bengali speakers, ideally ASHA workers
"Already registered" message	Reveals that a number is in the registry	Acceptable for V1; revisit if misused
Legal review	Personal data falls under India's DPDP Act	Review consent text and privacy page before scaling
Approximate pincode coordinates	Some distances are off by several km	Refresh from a newer India Post release
Lost private links	Can't be re-sent	Admin deletes the record; donor registers again
Free-plan limits	Supabase 500 MB database, limited backups; Netlify free usage limits	Export CSV backups; upgrade if usage grows
Name "Rakt Setu"	May already be used elsewhere	Check before printing posters

Ideas for V2

- WhatsApp number verification for new donors
- A volunteer mode for ASHA workers to see the donors they registered
- Per-district dashboard: donors by blood group, requests per week
- SMS fallback for donors without WhatsApp
- Show blood-bank stock from e-RaktKosh if an official data feed becomes available

References

- NBTC Blood Donor Selection Guidelines 2017: donation gaps, age, weight
- Blood donation: latest guidelines by MoHFW (M3 India): Hb, pulse, BP, deferral periods
- Blood donation in India (Wikipedia)
- e-RaktKosh: national blood-bank stock portal
- Supabase: do free projects pause when inactive?
- IndiaPIN: All India Pincode Directory with latitude and longitude
- All India Pincode Directory (data.gov.in mirror)
- Code: rakt-setu and Blood-Donation-Site